

BÀI 9. BIẾN ĐỔI ĐƠN GIẢN BIỂU THỨC CHỨA CĂN BẬC HAI**1) Khử mẫu của biểu thức lấy căn**

$$\frac{A}{\sqrt{B}} = \frac{A \cdot \sqrt{B}}{\sqrt{B} \cdot \sqrt{B}} = \frac{A\sqrt{B}}{B} \quad (B > 0)$$

Ví dụ: $\frac{3}{2\sqrt{6}} = \frac{3 \cdot \sqrt{6}}{2\sqrt{6} \cdot \sqrt{6}} = \frac{3\sqrt{6}}{2 \cdot 6} = \frac{\sqrt{6}}{4}$

$$\sqrt{\frac{A}{B}} = \sqrt{\frac{A \cdot B}{B \cdot B}} = \frac{\sqrt{A \cdot B}}{\sqrt{B^2}} = \frac{\sqrt{A \cdot B}}{B} \quad (A \geq 0, B > 0)$$

Ví dụ: $\sqrt{\frac{3}{5}} = \sqrt{\frac{3 \cdot 5}{5 \cdot 5}} = \frac{\sqrt{15}}{\sqrt{5^2}} = \frac{\sqrt{15}}{5}$

2) Trục căn thức ở mẫu

Sử dụng hằng đẳng thức: $(\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b}) = (\sqrt{a})^2 - (\sqrt{b})^2 = a - b$

Khi đó, ta nói: $\sqrt{a} - \sqrt{b}$ và $\sqrt{a} + \sqrt{b}$ là hai biểu thức liên hợp với nhau.

$$1) \frac{C}{\sqrt{A} + B} = \frac{C \cdot (\sqrt{A} - B)}{(\sqrt{A} + B) \cdot (\sqrt{A} - B)} = \frac{C \cdot (\sqrt{A} - B)}{(\sqrt{A})^2 - B^2} = \frac{C \cdot (\sqrt{A} - B)}{A - B^2} \quad (A \neq B^2, A \geq 0)$$

$$2) \frac{C + D}{\sqrt{A} - \sqrt{B}} = \frac{(C + D) \cdot (\sqrt{A} + \sqrt{B})}{(\sqrt{A} - \sqrt{B}) \cdot (\sqrt{A} + \sqrt{B})} = \frac{(C + D) \cdot (\sqrt{A} + \sqrt{B})}{(\sqrt{A})^2 - (\sqrt{B})^2}$$

$$= \frac{(C + D) \cdot (\sqrt{A} + \sqrt{B})}{A - B} \quad (A > 0, B > 0, A \neq B)$$

Ví dụ:

$$\frac{\sqrt{8} - \sqrt{5}}{2\sqrt{2} + \sqrt{5}} = \frac{(\sqrt{8} - \sqrt{5}) \cdot (2\sqrt{2} - \sqrt{5})}{(2\sqrt{2} + \sqrt{5}) \cdot (2\sqrt{2} - \sqrt{5})} = \frac{8 - 2\sqrt{10} - 2\sqrt{10} + 5}{(2\sqrt{2})^2 - (\sqrt{5})^2} = \frac{13 - 4\sqrt{10}}{3}$$

Bài 1. Trục căn ở mẫu của các biểu thức sau:

1) $3\sqrt{\frac{1}{3}}$

2) $\frac{6}{\sqrt{2}}$

3) $\frac{15}{\sqrt{5}}$

4) $\frac{5}{\sqrt{10}}$

Bài 2. Trục căn ở mẫu bằng cách đặt nhân tử chung:

1) $\frac{\sqrt{21} - \sqrt{7}}{\sqrt{3} - 1}$

2) $\frac{3 - \sqrt{3}}{\sqrt{3}}$

3) $\frac{\sqrt{10} - \sqrt{2}}{1 - \sqrt{5}}$

4) $\frac{\sqrt{6} - \sqrt{2}}{3 - 3\sqrt{3}}$

5) $\frac{2\sqrt{5} + 5\sqrt{2}}{2\sqrt{10}}$

6) $\frac{2\sqrt{3} - 3\sqrt{2}}{\sqrt{3} - \sqrt{2}}$

Bài 3. Trục căn ở mẫu bằng cách nhân lượng liên hợp:

1) $\frac{7}{3 - \sqrt{2}} - \frac{1}{\sqrt{2} + 1}$

2) $\frac{1}{2 + \sqrt{3}} + \frac{2}{\sqrt{3} - 1}$

3) $\frac{4}{3 - \sqrt{5}} - \frac{1}{\sqrt{5} - 2}$

4) $\frac{1}{2 - \sqrt{5}} - \frac{1}{2 + \sqrt{5}}$

$$5) \frac{4}{3-\sqrt{5}} - \frac{8}{1+\sqrt{5}} + \frac{5}{\sqrt{5}}$$

$$7) \frac{-1}{3+2\sqrt{3}} + \frac{1}{3-2\sqrt{3}} + \frac{2}{\sqrt{3}}$$

$$9) \frac{15}{\sqrt{3}} - \frac{\sqrt{3}-2}{\sqrt{3}+2} - \frac{1}{2-\sqrt{3}}$$

$$11) \frac{2}{\sqrt{5}+2} + \frac{4}{\sqrt{5}-1} - 3\sqrt{5}$$

$$13) \frac{2}{\sqrt{3}+1} + \sqrt{\frac{2}{2-\sqrt{3}}}$$

$$15) \sqrt{12+6\sqrt{3}} - \sqrt{\frac{2}{2+\sqrt{3}}}$$

$$17) \frac{\sqrt{2}}{\sqrt{5}-2} - \frac{\sqrt{2}}{\sqrt{5}+2} + \frac{7}{3-\sqrt{2}}$$

$$6) \frac{4}{3+\sqrt{5}} - \frac{8}{1+\sqrt{5}} + \frac{15}{\sqrt{5}}$$

$$8) \frac{2}{1-\sqrt{3}} - \frac{4}{1+\sqrt{3}} + \frac{9}{\sqrt{3}}$$

$$10) \frac{4}{\sqrt{3}-1} - \frac{\sqrt{3}-1}{\sqrt{3}+1}$$

$$12) \frac{5}{\sqrt{5}+\sqrt{2}} - \frac{2}{\sqrt{7}-\sqrt{5}} + \frac{3}{\sqrt{5}+\sqrt{2}}$$

$$14) \sqrt{\frac{2-\sqrt{3}}{2+\sqrt{3}}} + \sqrt{\frac{2+\sqrt{3}}{2-\sqrt{3}}}$$

$$16) \frac{6}{\sqrt{7}+2} + \sqrt{\frac{2}{8+3\sqrt{7}}}$$

$$18) \sqrt{\frac{2}{4-\sqrt{7}}} \cdot \frac{3}{\sqrt{7}+1}$$

Đs: 1) 4; 2) 3; 3) 1; 4) $2\sqrt{5}$; 5) 5; 6) 5; 7) $-\frac{2\sqrt{3}}{3}$; 8) 1; 9) 5; 10) $3\sqrt{3}$;

11) -3; 12) $-2\sqrt{2}$; 13) $2\sqrt{3}$; 14) 4; 15) 4; 16) $\sqrt{7}-1$; 17) $5\sqrt{2}+3$; 18) 1.

Bài 4. Thực hiện phép tính:

$$1) \frac{\sqrt{15}-\sqrt{12}}{\sqrt{5}-2} - \frac{1}{2-\sqrt{3}}$$

$$3) \frac{5\sqrt{2}-2\sqrt{5}}{\sqrt{5}-\sqrt{2}} + \frac{6}{2-\sqrt{10}}$$

$$5) \frac{\sqrt{10}-\sqrt{6}}{\sqrt{5}-\sqrt{3}} - \frac{7}{\sqrt{7}} + \frac{2}{\sqrt{7}-\sqrt{5}}$$

$$7) \frac{\sqrt{27}-3\sqrt{2}}{\sqrt{3}-\sqrt{2}} + \frac{12}{3+\sqrt{3}} + \frac{6}{\sqrt{3}}$$

$$9) \frac{\sqrt{6}-\sqrt{2}}{1-\sqrt{3}} + \frac{3}{2\sqrt{2}+3} - \frac{4}{\sqrt{2}}$$

$$2) \frac{5\sqrt{3}-3\sqrt{5}}{\sqrt{5}-\sqrt{3}} + \frac{1}{4+\sqrt{15}}$$

$$4) \frac{5\sqrt{3}-3\sqrt{5}}{\sqrt{5}-\sqrt{3}} - \frac{2}{4-\sqrt{15}} + 3\sqrt{\frac{5}{3}}$$

$$6) \frac{\sqrt{10}-\sqrt{2}}{\sqrt{5}-1} - \frac{7}{\sqrt{7}} + \frac{5}{\sqrt{7}+\sqrt{2}}$$

$$8) \left(\frac{\sqrt{6}+\sqrt{2}}{1+\sqrt{3}} + \frac{5}{\sqrt{5}} \right) : \frac{1}{\sqrt{5}-\sqrt{2}}$$

$$10) \frac{2\sqrt{5}-5\sqrt{2}}{\sqrt{5}-\sqrt{2}} - \frac{2}{\sqrt{10}+3} + \frac{20}{\sqrt{10}}$$

Bài 5. Rút gọn với các biểu thức sau đã có nghĩa:

$$1) \frac{x-4}{\sqrt{x}+2}$$

$$3) \frac{\sqrt{x}}{\sqrt{x}-2} + \frac{\sqrt{x}-1}{\sqrt{x}+2} + \frac{\sqrt{x}-10}{x-4}$$

$$5) \frac{1}{x-\sqrt{x}} + \frac{1}{x+\sqrt{x}} + \frac{2\sqrt{x}}{1-x}$$

$$2) \frac{x-\sqrt{x}-2\sqrt{x}+2}{x-1}$$

$$4) \frac{\sqrt{x}}{\sqrt{x}+3} + \frac{2\sqrt{x}}{\sqrt{x}-3} + \frac{3x+9}{9-x}$$

$$6) \left(\frac{1}{x-\sqrt{x}} - \frac{\sqrt{x}}{1-\sqrt{x}} \right) : \frac{1+\sqrt{x}}{x-1}$$

❖ Chủ đề: GIẢI PHƯƠNG TRÌNH CHỨA CĂN

Bài 1. Giải phương trình dạng $\sqrt{A^2} = B$

1) $\sqrt{(2x+1)^2} = 5$

2) $\sqrt{4(1-2x)^2} = 6$

3) $\sqrt{9x^2 - 6x + 1} - 2 = 0$

4) $\sqrt{4x^2 - 12x + 9} - 3 = 0$

5) $\sqrt{4 - 12x + 9x^2} - 1 = 1$

6) $\sqrt{x^2 - 8x + 16} - 2 = 1$

7) $5 - \sqrt{x^2 + 6x + 9} = 2$

8) $1 - \sqrt{9x^2 + 30x + 25} = \frac{1}{2}$

9) $\sqrt{x^2 + 4x + 4} = 2x - 1$

10) $\sqrt{9 - 6x + x^2} = x$

11) $x + \sqrt{x^2 + 4x + 4} = 6$

12) $\sqrt{x^2 - 6x + 9} = x - 2$

13) $2x - \sqrt{x^2 + 2x + 1} = 1$

14) $x - \sqrt{x^2 + 4x + 4} = 1$

15) $\sqrt{1 - 2x + x^2} = \sqrt{9 + 6x + x^2}$

16) $\sqrt{9x^2 - 6x + 1} = \sqrt{4x^2 - 4x + 1}$

17) $\sqrt{25x^2 - 10x + 1} = \sqrt{9x^2 - 6x + 1}$

18) $\sqrt{9x^2 - 12x + 4} = \sqrt{4x^2 - 4x + 1}$

19) $\sqrt{x^2 + x + \frac{1}{4}} = \sqrt{4x^2 + 12x + 9}$

20) $\sqrt{9x^2 - 12x + 4} = \sqrt{4x^2 - 12x + 9}$

Đs: 1) $\{2, -3\}$; 2) $\{-1, 2\}$; 3) $\left\{1, -\frac{1}{3}\right\}$; 4) $\{3, 0\}$; 5) $\left\{0, \frac{4}{3}\right\}$; 6) $\{7, 1\}$; 7) $\{0, -6\}$; 8) $\left\{-\frac{3}{2}, -\frac{11}{6}\right\}$; 9) $\{3\}$; 10) $\left\{\frac{3}{2}\right\}$; 11) $\{2\}$; 12) $\left\{\frac{5}{2}\right\}$; 13) $\{2\}$; 14) $\emptyset$; 15) $\{-1\}$; 16) $\left\{0, \frac{2}{5}\right\}$;17) $\left\{0, \frac{1}{4}\right\}$; 18) $\left\{1, \frac{3}{5}\right\}$; 19) $\left\{-\frac{5}{2}, -\frac{7}{6}\right\}$; 20) $\{-1\}$.**Bài 2.** Giải phương trình dạng $\sqrt{A} = B, \sqrt{A} = \sqrt{B}$

1) $\sqrt{4x-1} = 5$

2) $\sqrt{2x+1} = x+1$

3) $\sqrt{4x^2-3} = 2x-1$

4) $\sqrt{x^2-1} = x-2$

5) $\sqrt{3x+5} = \sqrt{9}$

6) $\sqrt{5+2x} = \sqrt{3}$

7) $\sqrt{2x+8} = \sqrt{3x}$

8) $\sqrt{4x-2} = \sqrt{x+1}$

9) $\sqrt{4x+12} - \sqrt{x+3} = 2$

10) $\sqrt{x-1} + \sqrt{4x-4} = 6$

11) $3\sqrt{20x-12} - \frac{2}{3}\sqrt{45x-27} = 8$

12) $3\sqrt{9x-45} - \sqrt{4x-20} = 21$

13) $5\sqrt{x-2} = 10 + \sqrt{9x-18}$

14) $\sqrt{9x+9} - 2\sqrt{\frac{x+1}{4}} = 4$

15) $\sqrt{4x-8} - 7\sqrt{\frac{x-2}{49}} = 5$

16) $6\sqrt{\frac{x-1}{9}} + \sqrt{16x-16} = 6$

17) $\sqrt{9x-27} + \frac{1}{2}\sqrt{4x-12} - 9\sqrt{\frac{x-3}{9}} = 2$

18) $2\sqrt{x-1} + 4\sqrt{\frac{x-1}{4}} - \sqrt{9x-9} = 2$

$$19) 16\sqrt{\frac{2x-1}{4}} - 2\sqrt{18x-9} = 4$$

$$20) 2\sqrt{9x-45} - \frac{5}{2}\sqrt{4x-20} = 5$$

ĐS.1) $\left\{\frac{13}{2}\right\}$; 2) $\{0\}$; 3) $\{1\}$; 4) $\emptyset$; 5) $\left\{\frac{4}{3}\right\}$; 6) $\{-1\}$; 7) $\{8\}$; 8) $\{1\}$; 9) $\{1\}$; 10) $\{5\}$;

11) $\left\{\frac{7}{5}\right\}$;

12) $\{14\}$; 13) $\{27\}$; 14) $\{3\}$; 15) $\{27\}$; 16) $\{2\}$; 17) $\{7\}$; 18) $\{5\}$; 19) $\left\{\frac{5}{2}\right\}$; 20) $\{30\}$.